Paleomagnetism of the Whitestone Anorthosite and Diorite, the Grenville Polar Track, and Relative Motions of the Laurentian and Baltic Shields[1]

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The Whitestone anorthosite and diorite are situated in the Grenville Structural Province north of Parry Sound, Ontario. They are intruded into sediments and igneous rocks, the whole being metamorphosed to amphibolite facies. Aside from soft magnetizations due to the present field four magnetizations are present, two owing to hematite, the third mainly to magnetite, and a fourth of uncertain source. It is argued that these are thermoremanent magnetizations acquired during very slow cooling following regional metamorphism in the interval 1100 to 1000 m.y. A single-stage cooling model based on Neel's single domain theory is developed, which suggests that the hematite magnetizations were acquired during slow cooling at about 240 °C and the magnetite magnetizations at about 200 °C. The poles from Whitestone rocks fall among a group of poles from elsewhere in the Grenville Province. There are serious problems in integrating these Grenville poles with those from other parts of the Canadian Shield, and three possible ways of relating them are evaluated. Poles from Grenville-type rocks from the Baltic Shield are near to the Grenville poles after correction is made for the late Phanerozoic opening of the Atlantic, showing that the relative positions of Laurentian and Baltic Shields before and after the Caledonian orogeny were very similar. There is however a small but significant difference, and this is attributed to Caledonian diastrophism.

Les anorthosites et diorites de Withestone sont situés dans la Province Structurale Grenville au nord de Parry Sound, Ontario. Ils sont intrusifs dans des sédiments et des roches ignées, l'ensemble étant métamorphisé au faciès amphibolite. En excluant la faible magnétisation due au champ actuel, quatre magnétisations sont présentes, deux provenant de l'hématite, la troisième surtout de la magnétite, et la quatrième de source incertaine. Il est soutenu que ces magnétisations thermorémanentes ont été acquises durant un refroidissement très lent après un métamorphisme régional situé dans l'intervalle 1100 à 1000 m.a. Un modèle de refroidissement en une seule étape s'appuyant sur la théorie du domaine unique de Neel, est développé et suggère que la magnétisation de l'hématite a été acquise durant un refroidissement lent à environ 240 °C et la magnétisation de la magnétite à environ 200 °C. Les pôles établis à partir des roches Withestone tombent dans un groupe de pôles de la Province Grenville. L'intégration des pôles du Grenville avec ceux d'autres parties du Bouclier Canadien pose de sérieux problèmes et trois façons possibles de les mettre en relation sont évaluées. Les pôles de roches de type Grenville provenant du Bouclier Baltique sont près des pôles du Grenville, après une correction pour l'ouverture au Phanérozoïque supérieur de l'Atlantique et montrent que les positions relatives des Boucliers Laurentien et Baltique avant et après l'orogène calédonienne sont très similaires. Il y a cependant une différence petite mais significative, et elle est attribuée au diastrophisme Calédonien.

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Introduction

Paleomagnetic results from amphibolite grade metamorphic rocks of the Grenville Structural Province near Parry Sound, Ontario are described (Fig. 1), and pole positions calculated. These poles along with those other parts of the Grenville Province form a broad zone called the Grenville Track. There are major problems in integrating Grenville poles with those from other parts of the Canadian Shield and this problem is discussed. The Grenville poles are then compared with those from comparable terrain in Scandinavia and a tentative estimate of the relative positions of the Baltic and Canadian Shields prior to the Caledonian orogeny obtained.

It has been customary in paleomagnetic work to select specimens from essentially unaltered rocks, in the belief that the remanent magnetization provides a record of the geomagnetic field at the time of formation. The Grenville Province consists for the most part of high-grade meta-

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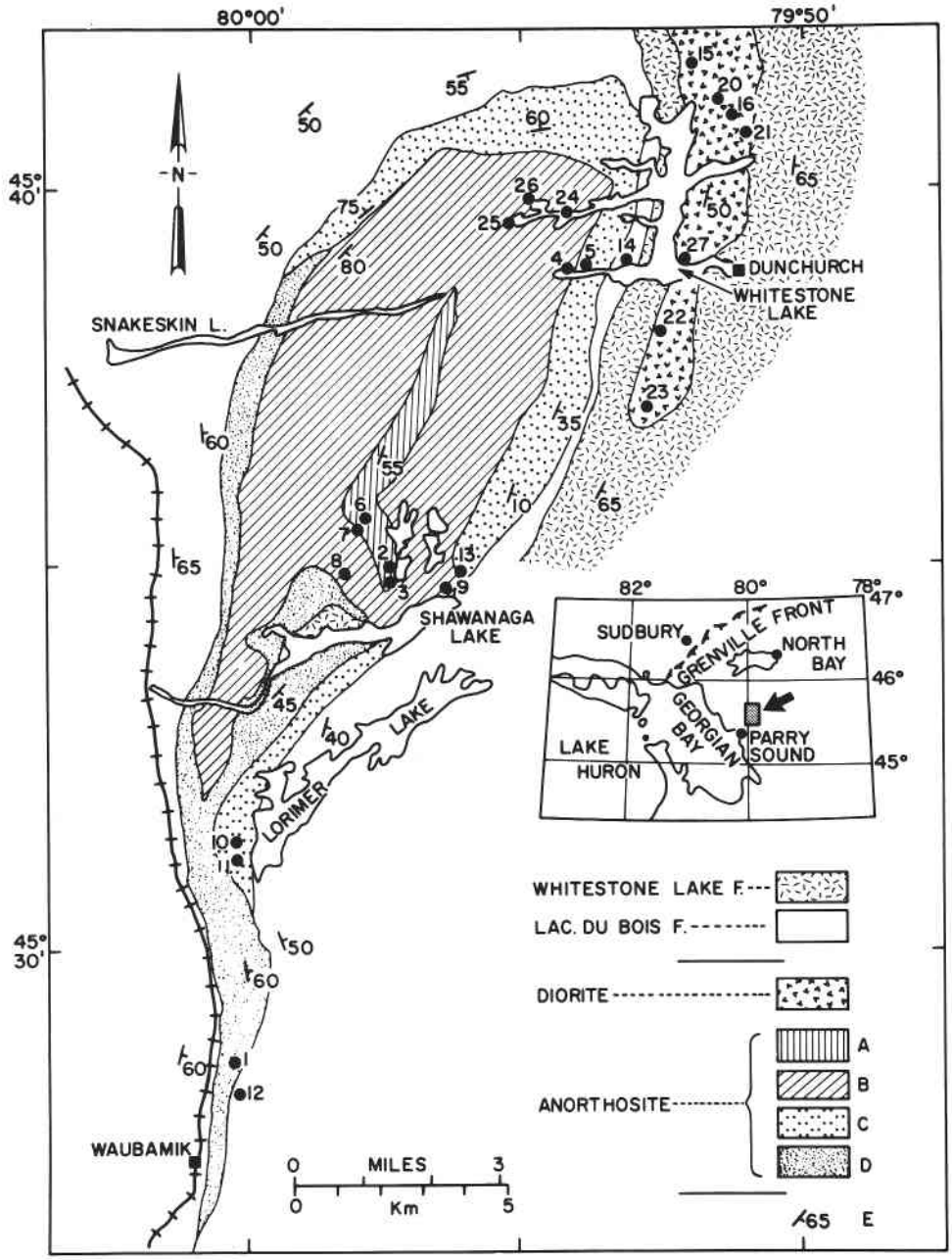


FIG. 1. Sketch-map of Whitestone area. Geology from Mason (1969). The sampling sites are shown by bold dots. Their grid references are given in Table 1. Two localities (18, 19) are from the diorite 3 km to the north of site 15. One locality (17) in amphibolite from a road cutting on highway 69 at Parry Sound is not marked. For the explanation of A, B, C and D, see text. E are foliations.

morphic rocks in which the original magnetization has been lost during metamorphism and replaced subsequently. There are likely to be two main types of natural remanent magnetization in such terrain, thermoremanent magnetization (t.r.m.) due to cooling following regional metamorphism or subsequent reheating, and chemical remanent magnetization (c.r.m.) due to formation of new minerals at temperatures below their blocking temperatures. But the

situation is different from that in conventional paleomagnetic work because the time scale of processes is very much longer. For example during very slow cooling following regional metamorphism, magnetizations may be acquired not near the Curie temperature as in rapidly cooled volcanic rock, but at much lower temperatures, and it is to be expected that the magnetization will relate to the later stages of slow regional uplift and cooling. A qualitative model is proposed.

Geology of the Anorthosite

The Whitestone anorthosite and diorite are conformably emplaced into gneisses and marbles of the Whitestone Lake Formation, and into migmatites and amphibolites of the Lac du Bois Formation, which lie in a belt of metasedimentary and metavolcanic rocks trending NE from Parry Sound to near North Bay (Fig. 1). The anorthosite is a NE-trending, tadpole-shaped body, covering approximately 130 km^2. The diorite has an area of 14 km^2. The region is metamorphosed to amphibolite grade.

The area was first mapped by Lacy (1960). More recently, Mason (1969) completed a detailed mapping (4″ = 1 mile) and petrochemical study of the anorthosite and adjacent country rock. He recognized two primary facies of the anorthosite, the *glomeroporphyritic* facies (*B* in Fig. 1) in which clinopyroxene occurs in clots charged with plagioclase grains, and the *porphyritic* facies in which the pyroxene is more evenly distributed. He also recognized three superimposed secondary facies: the *green feldspar* facies (*A*), a *marginal foliated garnet-bearing facies* (*C*), and the *marginal foliated epidote* facies (*D*). Traced towards the margin from facies *B*, the foliation develops, by the elongation of the glomerules, into discontinuous bands and streaks. There is a sympathetic development of garnet and/or epidote, and a change in color of the feldspar from blue-grey to milky-white. There is no evidence of minor folding or lineation in the plane of the foliation, but such features are common in the country rock.

Mason found the following progressive changes from the foliated margins to the massive core: the plagioclase content increases from 60% to over 90%, the amphibole/amphibole + pyroxene ratios decrease from 0.85 to <0.20, garnet decreases from >5% to >0.5%, and epidote decreases from >1% to near zero. At the margins there is evidence of cataclasis (grain margin granulation, optically strained grains, bent twin lamallae), whereas in the core, the plagioclase shows a polygonal mosaic of unstrained, equidimensional grains. A thin contact thermal and metasomatic aureole surrounds the anorthosite.

Mason considers the anorthosite to be a sill, overturned to the west and plunging NNE. He believes that it was intruded as one of the last events in the area mainly because the anorthosite is much less foliated than the country rock. He argues that the recrystallization of the plagioclase, the development of the foliation, and the growth of garnet and epidote, occurred during post-magmatic cooling. He believes that the uralitization of pyroxene and scapolitization of plagioclase are also part of this 'autometamorphic' event. During crystallization high P_{H_2O} and P_{O_2} conditions prevailed, and this may explain the absence of sulfides, the predominance of hemo-ilmenite (see below) and the absence of ulvospinel in the oxides, the development of amphiboles and scapolite, and the presence of the metasomatic-metamorphic aureole. However it is also possible to argue that the intrusion of the anorthosite and its metamorphism were widely separated in time. Since the metamorphism (whether it occurred very soon or long after emplacement) was intense, there can have been no magnetic memory of prior events, so from the standpoint of the present work the distinction between these two possibilities is not important.

Geochronology

There are no isotopic ages from the anorthosite or the diorite. However, their age can be bracketed by using data from this part of the Grenville Province.

Mineral ages by the K/A, Rb/Sr and U/Pb methods all suggest that a metamorphic event occurred about 1000 m.y. ago, and was of sufficient intensity that isotopic reequilibration occurred between adjacent mineral grains. Whole-rock Rb/Sr data show that much older intrusive and metamorphic events also occurred.

K/A ages on muscovite and biotite in a young cross-cutting pegmatite dike, at Blackstone Lake, 40 km SE of the Whitestone anorthosite, give ages of 930–990 m.y. (Lowden 1960). Eighty kilometers to the NW, a biotite – whole-rock – apatite Rb/Sr age of 900–1000 m.y. has been obtained from the French River granite (Krogh

et al. 1967), while adjacent units have a plagioclase – whole-rock age of 800–900 m.y. (Krogh and Davis 1969). A paragneiss unit at Pointe au Baril, 30 km to the NW, yields plagioclase – microcline – whole-rock and biotite – whole-rock Rb/Sr ages of 940 and 890 m.y. respectively. In addition 2–3 mm wide sheets within 2–3 cm wide layers in the paragneiss gave ages of approximately 950 m.y. (Krogh and Davis 1969, 1970). Seventy kilometers to the SW, Rb/Sr apatite – whole-rock and biotite – whole-rock ages of 700–800 m.y. have been determined in the Muskoka granite (Krogh *et al.* 1967). U/Pb studies on two sphenes from the French River granite gave concordant ages of about 1100 m.y. (Tilton and Gruenfelder 1968), somewhat older than the Rb/Sr mineral ages from the same body. In the paragneiss unit at French River, Krogh and Davis (1972) found that, unlike the Rb/Sr whole-rock data (see below), zircons gave no evidence of older source material, and indeed suggest that the U/Pb system was disturbed at not more than 1300 m.y. ago. However, the data are inconclusive. They also studied zircons from two pegmatite pods, which appear to have formed late in the area's history, and obtained ages of 1025 m.y. Finally Lowden (1960) reports a uraninite U/Pb age of 1005 m.y. from the Blackstone Lake pegmatites. In short, the mineral data show that a significant metamorphic event occurred between 1000–1100 m.y. ago with a possible event as early as 1300 m.y. Krogh and Davis (1972) suggest that the U/Pb ages reflect the closing stages of metamorphism and folding, whereas the K/A and Rb/Sr ages reflect final cooling.

Rb/Sr whole-rock studies 60–150 km south of the Grenville Front show at least three periods of granitic intrusive activity, at 1700, 1500 and 1300 m.y. (Davis *et al.* 1966; Krogh *et al.* 1967). The study of banded paragneiss units at French River and at Pointe au Baril showed that the 2–3 cm wide layers and the whole-rock were isotopically homogeneous 1800 m.y. ago, implying that initial formation from the sedimentary parent occurred at this time. As mentioned above the 2–3 mm wide sheets reequilibrated at 1000 m.y. (Krogh *et al.* 1967; Krogh and Davis 1968, 1969, 1970).

Preliminary work on a paragneiss unit at Parry Sound, 20 km from the anorthosite (the closest available isotopic data), give a Rb/Sr age of about 1170 m.y. (Krogh and Davis 1968). By analogy with the work at French River and Pointe au Baril this age indicates the time of formation of paragneiss from parent sediment, that is, the time of the main metamorphism.

The age of the anorthosite and diorite may be bracketted as follows. Field relationships show that the anorthosite was intruded before the end of the metamorphic event. Hence its younger age limit is probably about 1000–1025 m.y. The intrusion imparted a thermal and metasomatic aureole on the country rocks, which are presumably of the same ages as the Parry Sound rocks (Mason 1969; Mummery 1973). Hence the maximum age is probably about 1170 m.y.

Methods

Sampling was confined to the eastern half of the anorthosite, access to the western part being difficult. A general coverage of the diorite was obtained (Fig. 1). Samples were cored in the field and oriented by magnetic and sun compass (Haaf and Winsink 1968) to an accuracy of about 3°. Remanent magnetization was measured on a Schonstedt SSM1 magnetometer. Thermal and alternating field (a.f.) demagnetization was carried out using the apparatus described by Roy *et al.* (1969, 1971, 1973). Fields in the demagnetizing apparatus were less than 2 gammas, and specimens were tumbled about 3 axes. Generally 2 specimens were cut from each core. An attempt was made to determine stable-end points for one specimen from each sample. The second specimen of each pair was then demagnetized in the end-point field. The results from pairs of specimens were averaged and these averages are the basic statistical unit. Stable end-points were found in 83 (17 sites) of the 168 samples (27 sites) collected. At the other sites no stable end-points were found and the directions were random. Thermal demagnetization of n.r.m. was carried out on 16 specimens, but in order to determine the blocking temperature of the high remanent coercive force (h.c.f.) component, 13 specimens were first demagnetized (a.f.) in their stable-end point fields, and subsequently demagnetized in that field between each heating step in order to ensure the removal of spurious components that might be acquired during thermal demagnetization.

The sampling sites are shown in Fig. 1. Samples of amphibolite near Parry Sound, from sites 12 and 14 in the contact aureole, sites 1 and 5 in the highly foliated margin, and site 8, which

was somewhat weathered, gave no stable endpoints. The reason why there is no stable magnetization at these 10 sites is not known and no further studies of these samples were made.

Remanence of Anorthosite

The anorthosite has three magnetizations. *WW* is the dominant magnetization of the oxide segregations, *WX* is a subordinate magnetization found in some of the oxide segregations, and *WY* is the magnetization of the massive anorthosite.

WW Magnetization

Samples from the garnet-bearing eastern margin of the anorthosite commonly contain oxide segregations. Kretschmar and McNutt (1971) found that in these segregations hemo-ilmenite (bulk composition Hem_{10} $Ilmen_{30}$) predominates, with pure magnetite (<3.5 wt% ulvospinel in solid solution) as a minor phase. Their chemical composition and phase relationships indicate a final temperature of equilibration of approximately 550–600 °C using the experimental results of Buddington and Linsley (1964). The hemo-ilmenite occurs as aggregates of grains 1–5 mm in size containing exsolved lamellae of hematite and ilmenite usually between 50 and 100 μm in thickness. Within the ilmenite (hematite) lamellae there are smaller hematite (ilmenite) lamellae ranging down in thickness to 0.5 μm or less. It is possible to explain these textures by a single cooling event, but multistage models are also possible. The magnetite occurs as euhedral homogeneous grains, 100–200 μm in diameter. They are either in contact with the hemo-ilmenite, or are surrounded by garnet. In approximately half of the sections of oxide segregations examined in the present study, magnetite was not observed.

The remanence of samples containing oxide segregations is generally very stable. The directions are commonly little changed after a.f. demagnetization in 2000 Oe, so that within any single specimen there is only a single high remanent coercive force (h.c.f.) magnetization, which is referred to as the *WW* magnetization. The median destructive fields generally exceed 1000 Oe (Fig. 2). The blocking temperatures of the *WW* magnetization range between 600–650 °C, indicating that it is due to fine-grained titanium-free magnetite and hematite, or to hematite alone (Fig. 3). The most probable

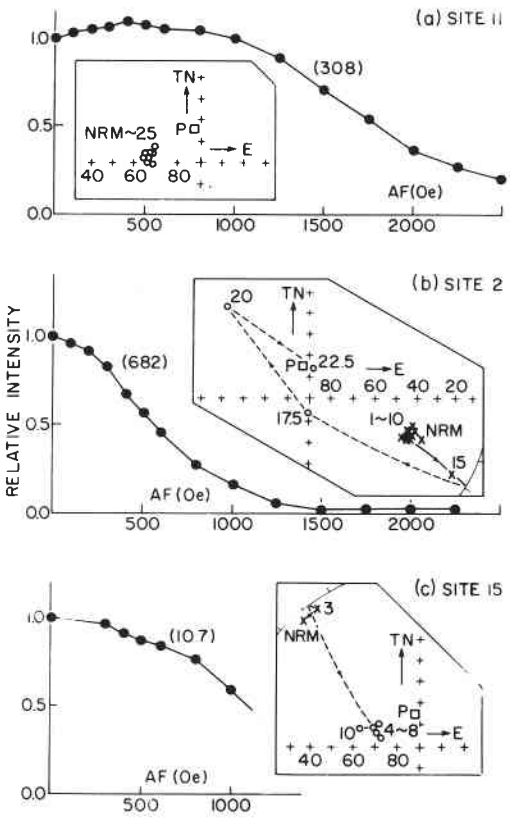


FIG. 2. Typical changes in directions and intensity during a.f. demagnetization. The intensity of n.r.m. is given in brackets (microgauss) above each curve. The directions are plotted in the insets which show parts of equal-area nets. Circles (crosses) denote upward (downward) inclinations. Each direction is labelled with the a.f. strength in Oe × 10^2. (a) *WW* magnetization in oxide segregation. (b) *WY* magnetization in massive anorthosite with small underlying *WW* magnetization. (c) *WZ* magnetization in diorite. *P* is the present field (down).

sources of the *WW* magnetization are the very fine hematite lamellae observed in polished sections. Although magnetite could contribute, it is unlikely that it does for two reasons. Firstly the observed magnetite grains are large and homogeneous, and are therefore unlikely to possess the observed high coercivities. Secondly, in half of the samples studied magnetite is not observable. However the possibility that the *WW* magnetization is partly due to small optically undetected magnetite grains cannot be entirely ruled out. The fine hematite lamellae must contain some ilmenite in solid-solution because the maximum blocking temperatures (650 °C) are less than that of pure hematite (675°C), although

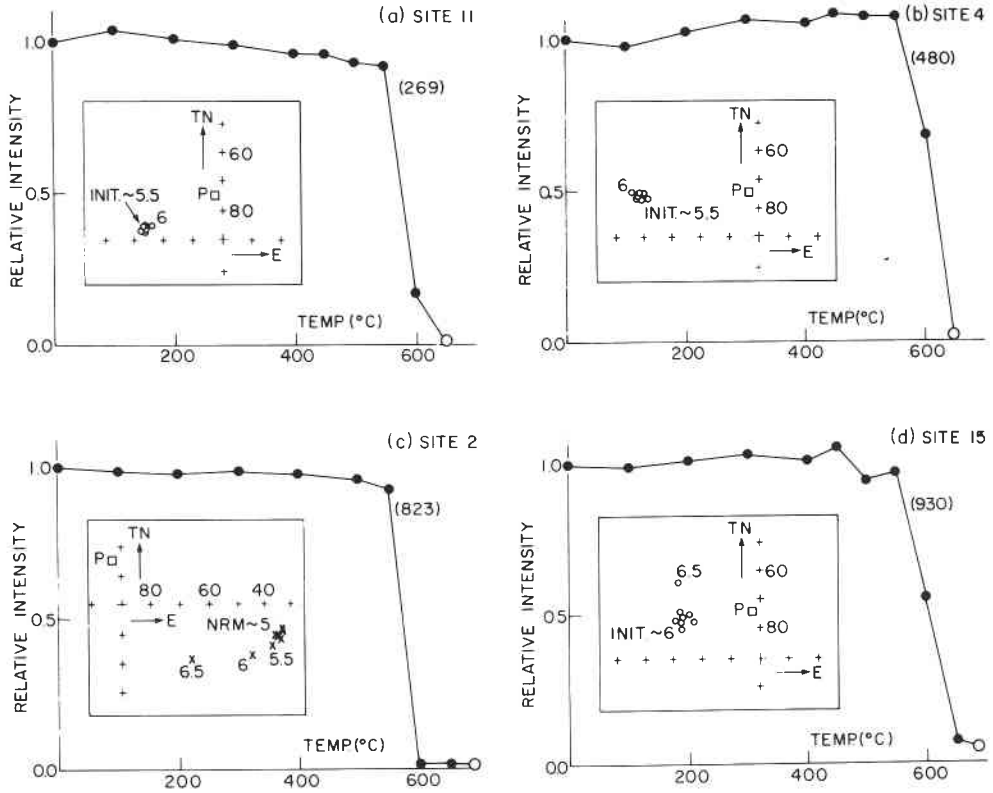


FIG. 3. Typical changes in direction and intensity during thermal demagnetization. Each direction is labelled with the appropriate temperature ($\times 10^2$ °C). (a) and (b) show the h.c.f. *WW* magnetization in oxide segregations. (c) *WY* magnetization in massive anorthosite. (d) the h.c.f. *WZ* magnetization in diorite. In (a), (b), and (d), specimens thermally demagnetized were first demagnetized in a.f. of 800, 600 and 600 Oe respectively, so the curves give the blocking temperatures of the stable component only. The n.r.m. intensities are given in brackets in microgauss except in (d) where the units are 10^{-8}. Other conventions as in Fig. 2.

distinctly higher than pure magnetite (560°C). Typical grouping of *WW* magnetization directions after a.f. cleaning is shown in Fig. 4(a). The mean site directions are listed in Table 1. The inclinations are upward.

WX Magnetization

Most samples with oxide segregations contain only the *WW* magnetization. At sites 4 and 9 however some samples have direction of n.r.m. toward the E or SE with downward inclination which migrate about 120° during demagnetization up to 1000 Oe (Fig. 5). The magnetization removed during each demagnetization step can be calculated by vector subtraction as described by Roy and Park (1974). For the steps up to 1000 Oe the removed magnetization in the specimen illustrated in Fig. 5 has the direction 132, +64. This is a medium remanent coercive force (m.c.f.) magnetization and is referred to as *WX*. For the steps from 1250 to 2250 Oe the removed magnetization has the direction 283, −74, which is the h.c.f. magnetization *WW*. The magnitudes of *WX* and *WW* are 81 and 11 microgauss respectively. Similar m.c.f. magnetizations coexisting with h.c.f. magnetizations are common in the Morin anorthosite (Irving *et al.* 1974), and this is a further illustration of the fact that for adequate removal of m.c.f. magnetization and disclosure of the h.c.f. magnetization in anorthositic rocks, fields as high as 1000 Oe are commonly required.

WY Magnetization

The massive anorthosite specimens from the center of the Whitestone body do not have oxide segregations. Their coercive forces are generally less than 1000 Oe, and fall in the range of m.c.f.

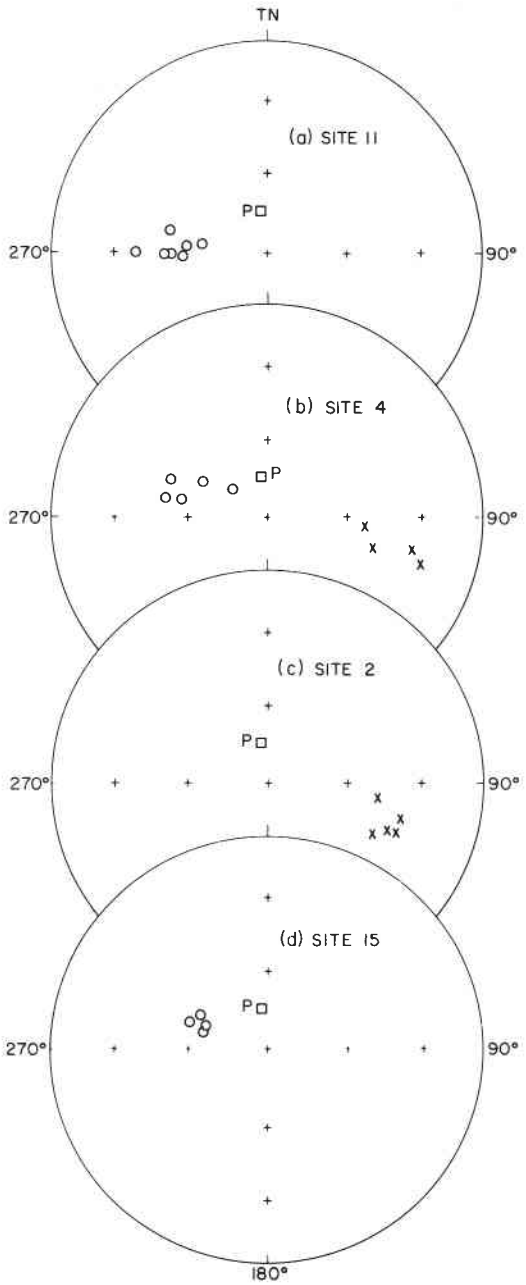


FIG. 4. Directions observed at four sites after a.f. cleaning. Circles (crosses) denote upward (downward) directions. *P* is the present field (downward). Sample means (2 specimens) are plotted. (a) Typical *WW* magnetization of specimens with oxide segregations. (b) Mixed *WW* (circles) magnetization in oxide segregations, and *WY* magnetization (crosses) in massive anorthosite within 9 m of one another at site 4. (c) Typical *WY* magnetization in massive anorthosite. (d) *WZ* magnetization in diorite.

magnetization, substantially less than those of the oxide segregations. Median destructive fields are typically about 500 Oe (Fig. 2). Blocking temperatures are in the range 550–590°C, but in some specimens a very small component is retained at 600°C (Fig. 3(c)). These properties indicate that *WY* magnetization is due predominantly to titanium-free magnetite with possibly a small contribution from hematite. Under reflected light the distinctive feature of these rocks is abundant very fine rods of magnetite in the plagioclase. The rods are a few microns long and have thicknesses of 2–0.5 μm or less, and are considered to be responsible for most of the *WY* magnetization. In addition there are occasional large (100 μm) magnetites which are in contact with hemo-ilmenite (100–10 μm) grains. Typical grouping of *WY* magnetization is illustrated in Fig. 4(c). The site directions are listed in Table 1.

In most specimens the magnetization consists of a single m.c.f. component, but in a few the directions remain consistent up to 1000 Oe and then migrate to the west with upward inclination (Fig. 2). The remaining component is very small (2%) and has a poorly defined direction *roughly* similar to the *WW* magnetization. This behavior is similar to that shown in Fig. 5 except that the *WW* magnetization is now only an insignificant part of the total.

At site 4 both *WW* (oxide segregations) and *WY* (massive anorthosite) magnetizations occur in different specimens (Fig. 4(b)). Also both *WW* and *WX* occur within the same specimen (Fig. 5). One core from site 4 has been divided into a specimen consisting largely of an oxide segregation which has *WW* magnetization only, and a specimen of massive anorthosite which has *WY* magnetization only. Thus there is a clear distinction in different sites, in cores from the same site, and even in different specimens from the same core, between oxide segregations whose magnetization (*WW*) very probably is due to titaniferous hematite, and massive anorthosite whose *WY* magnetization is predominantly due to magnetite. In addition, the former may or may not have a subordinate *WX* magnetization, and the latter may or may not have a subordinate *WW* magnetization.

Quartz Diorite: *WZ* Magnetization

The diorite has never been mapped in detail,

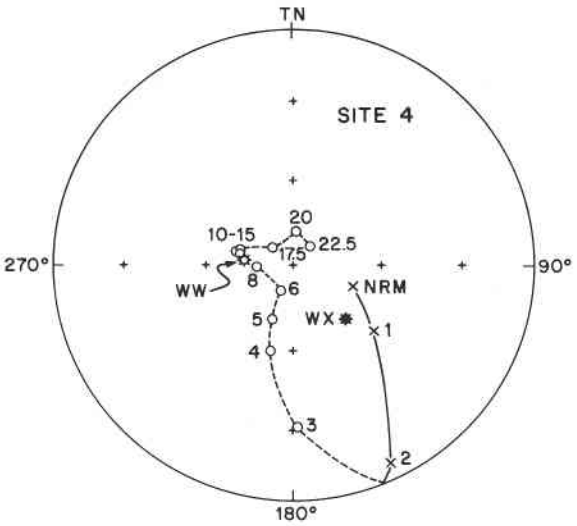


FIG. 5. *WX* and *WW* magnetization coexisting in same specimen. Circles (crosses) denote upward (downward) directions. See explanation in text.

and a brief summary of some thin-sections is given below. It is an amphibolite grade metamorphic rock.

Coherent magnetizations occur at only four (15, 16, 20, and 21) of the nine sites sampled. All four are from the massive central part of the body, and their mineralogy is similar with some modal variation. The felsic minerals are quartz, plagioclase (An_{35-40}) and minor orthoclase. Blue-green hornblende, biotite, and opaque oxides occur. There is some alteration of the plagioclase, but it is not extensive. Many of the amphiboles show oxide inclusions which have almost certainly formed by exsolution. Foliation is minor to absent, but there is some deformation because some plagioclase grains have bent or curved twin lamellae.

The remanence at these four sites consists of an h.c.f. magnetization with steep negative inclination. This is referred to as *WZ* magnetization. Low coercive force magnetizations of variable magnitude occur aligned roughly along the present field. Vector analysis showed that no superimposed m.c.f. component are present. The samples have excellent stable end-points (Fig. 2). Typical results after cleaning are plotted in Fig. 4(d). The site directions are listed in Table 1. Thermal demagnetization of the h.c.f. magnetization gave excellent square-shouldered curves with blocking temperatures of about 640°C. The magnetizations seems to be due entirely to hematite with some ilmentite in solid-solution, because only hemo-ilmenite grains are observed in polished sections. These range in

TABLE 1. Site directions

No.	UTMG	*M*	a.f.	*N*	*D*°, *I*°(*R*)
WW magnetization: oxide segregations					
04B	854555	388	6, 10	5	290, −59(4.92)
09	827476	208	4—10	7	296, −62(6.65)
10	777412	333	4—10	5	274, −54(4.89)
11	777411	310	8, 10	7	272, −54(6.93)
13	830479	105	4, 6	5	295, −49(4.95)
WY magnetization: massive anorthosite					
02	813480	416	8	5	108, +41(4.96)
03	813477	207	4	4	110, +36(3.97)
04A	854555	68	6	4	103, +41(3.91)
06	805489	304	8, 10	7	117, +44(6.92)
07	804487	176	6	5	100, +40(4.38)
24	853568	160	3	5	105, +40(4.71)
25	841565	50	3	4	74, +34(3.60)
26	845569	160	3, 6	5	120, +41(4.85)
WZ magnetization: diorite					
15	884605	7	6, 10	4	292, −62(3.99)
16	893593	23	6—10	4	286, −67(3.97)
20	891596	—	6	3	290, −71(2.95)
21	895589	—	4—8	4	298, −60(3.96)

Site numbers are given in the left-hand column. At site 4 both polarities occur. UTMG is the universal transverse Mercator grid reference, all numbers are prefixed by 18T. *M* is the mean n.r.m. intensity in microgauss. a.f. is the peak cleaning field in hundreds of oersted. *N* is the number of cores (2 specimens from each). D^0,I^0 is the mean site direction giving cores unit weight. *R* is the resultant.

size from 150 to 50 μm and contain two generations of exsolution as in the oxide segregations of the anorthosite.

The remaining five sites (18, 19, 22, 23 and 27) showed no stable end-points and the directions are random. Sites 18 and 19 at the northern end of the body (beyond limits of Fig. 1) have the same mineralogy as described above, but they show extensive recrystallization and are somewhat foliated. Sites 22 and 23 from the south end of the body are very different. Site 22 has an igneous texture somewhat cumulus in nature. Pleochroic, pink-green orthopyroxene fills the spaces around large felsic grains, which are predominantly plagioclase with minor quartz and orthoclase. The felsic grains are strained and plagioclase composition determinations are difficult. Associated with the orthopyroxene are minor amounts of clinopyroxene, probably augite. Site 22 is more basic than sites 15–21. Site 23 is a hornblende-plagioclase-garnet-quartz rock. It shows abundant evidence of recrystallization, displaying many polygonal grains of quartz and plagioclase, the latter showing simple twinning only. The garnet and hornblende occur together, while the opaques are always found associated with the garnet as inclusions. This rock is so different from the diorite as a whole that it may even be a separate body.

Remanence Anisotropy

The rocks sampled are often strongly foliated and the possibility that the magnetization was deflected from the ambient field is now considered. Samples showing typical *WW*,*WY* and *WZ* magnetizations were reoriented into their field positions in an electromagnet and a field of 20 000 Oe applied in the direction of the observed magnetization. The i.r.m. directions acquired were measured and the Fisher statistics calculated. Individual departures from the applied field are a measure of the effect of anisotropy. Departures ($\delta°$) of the mean from the applied field determine whether or not the individual departures are systematic. The specimens were then demagnetized in progressively stronger fields and the statics calculated at each step so that the behavior of progressively higher coercive force components can be determined (Irving and Park 1973).

The results are given in Table 2. In the oxide segregations the direction of i.r.m. (284.3, −51.4) does not differ significantly from the applied field (285, −56) and after demagnetization (284.0, −55.0) becomes closer to it. Clearly there is no reason to suppose that any part of the coercive force spectrum is affected by anistropy. The results from massive anorthosite with *WY* magnetization are more complex. Both the i.r.m. and the h.c.f. magnetization have declinations that agree with the applied field, but the inclinations are steeper by amounts that are marginally significant and that increase with increasing a.f. demagnetization. In the diorite the directions depart from the applied field by about 2° which is not significant, but the departures are still somewhat larger than for specimens with oxide segregations. In all specimens the dispersions owing to anisotropy are much less than the natural dispersion. Clearly anisotropy is not a *major* source either of dispersion or of systematic error, but it could have caused a systematic increase of inclination in *WY* specimens, and a systematic decrease in *WZ* specimens. These effects are small and marginally significant. Surprisingly, the *WW* magnetizations that are characteristic of the strongly foliated margins, are not affected systematically, whereas systematic effects, although small, appear to occur in the much less foliated massive diorite and massive central part of the Whitestone anorthosite.

Statistical Analysis

The usual statistics are given in Table 3. The *WW*, *WY* and *WZ* magnetizations are well-grouped. Only in *WW* is the between-site precision significant. No correction for anisotropy is applied to the *WW* magnetization, but the inclination for *WZ* has been increased by 2° and that for *WY* decreased by 7°. These are the mean deflections observed in the experiments of Table 2. These deviations are only marginally significant, but it is believed that they represent the best estimate of the small effects of anisotropy. The *WX* magnetization is found at only two sites, and is widely scattered, and the result gives only a rough indication of direction. The *WY* magnetization direction (irrespective of sign) is 13° and 22° different from the *WW* and *WZ* directions, and this is significant. The corresponding poles are plotted in Fig. 7.

Origin and Age of Magnetization

The mean blocking temperatures of the *WZ*, *WW* and *WY* magnetizations are 640, 620 and 570°C respectively. These results were obtained

TABLE 2. Remanence anisotropy

	M	$D°, I°$	k	$\alpha_{95}°$	$\delta°$
Specimens with oxide segregations and WW magnetization ($N = 5$)					
n.r.m.	229	352.4, +87.1	4	46.7	40
1000 Oe	34	280.4, −63.4	21	17.1	18
i.r.m. (285, −56)	23 400	284.3, −51.4	86	8.3	5
400 Oe	3090	283.2, −54.8	64	9.7	1
1000 Oe	1970	284.0, −55.0	45	11.5	1
Massive anorthosite specimens with WY magnetization ($N = 12$)					
n.r.m.	217	112.8, +43.3	25	8.8	16
800 Oe	49	109.5, +43.9	32	7.8	14
i.r.m. (104, +43)	4400	104.0, +47.0	472	2.0	4
400 Oe	1420	104.7, +50.2	506	1.9	7
800 Oe	600	104.9, +50.5	571	1.8	7
Diorite, specimens with WZ magnetization ($N = 5$)					
400 Oe	38	293.6, −65.2	65	9.5	10
i.r.m. (292, −65)	12 300	288.2, −62.9	261	4.7	2
400 Oe	4400	289.2, −63.2	256	4.8	2
600 Oe	4100	288.5, −62.9	297	4.4	2

Applied field directions are given in column 1 (in brackets). M is the mean intensity in microgauss. δ^0 is the departure of the mean direction from the applied field direction. Further explanation given in text.

TABLE 3. Summary of results

	WW	WX[3]	WY	WZ
By Sites[1]				
mean direction	285°, −56°	122°, +45°	104°, +43°	292°, −65°
direction corrected[4]	—	—	104°, +36°	292°, −67°
number	5	17	8	4
precision	96	3	45	223
error	8°	26°	8°	6°
Two-tier analysis[2]				
within-site precision	36	—	18	89
between-site precision	176	—	ns	ns
error	6°	—	7°	5°
Poles				
from mean direction	16°S, 156°E	01°S, 152°E	08°S, 165°E	21°S, 144°E
from corrected direction[4]	—	—	05°S, 168°E	22°S, 146°E
error (dm, dp)	11°, 8°	33°, 21°	10°, 6°	10°, 8°

NOTES: *ns* means not significant, $P = 0.05$.
All errors $P = 0.05$.
[1]By method of Fisher (1953).
[2]By method of Watson and Irving (1957).
[3]Samples unit weight, ($N = 17$).
[4]Corrected for anisotropy.

after heating at the specified temperature for about 5 min. When very slow cooling occurs the blocking temperatures are reduced, and may be estimated from the relationship assuming an assemblage of single-domain grains

$$T_1 \ln(3C\tau_1) = T_2 \ln(3C\tau_2)$$

T_1 and T_2 are the temperatures corresponding to relaxation times τ_1 and τ_2 (Neel 1949; Stacey 1963). The coefficient C is taken to be 10^{10} s^{-1} following Neel (1949). For a relaxation time of 10^5 y the blocking temperature of WZ, WW and WY magnetizations become 240, 230 and 200°C respectively. These values will increase (decrease) by about 20° as the relaxation time decreases (increases) by an order of magnitude. In these calculations the changes in the anistropy constant with temperature are assumed small, and

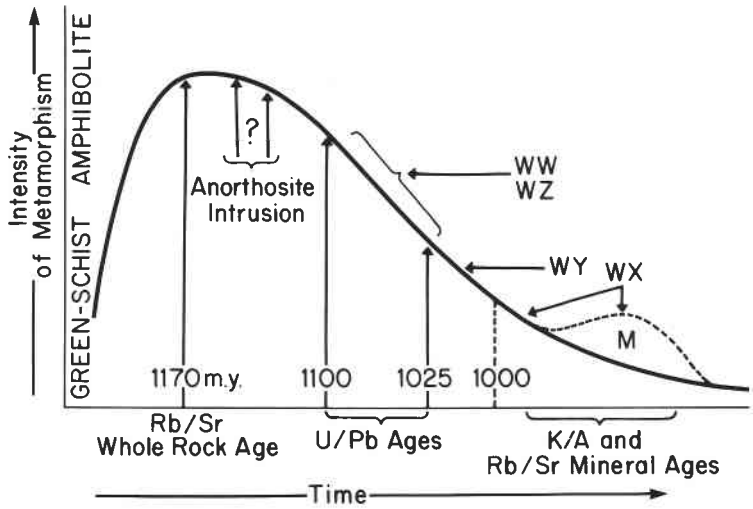


FIG. 6. Tentative model for interpretation of ages of magnetization of metamorphic rocks using the Whitestone results as an example. See explanation in text.

the results therefore provide only a rough guide to the temperatures at which magnetization is acquired. Nevertheless if, as seems likely, regional cooling even of quite small areas is to be measured in tens of thousands to millions of years, then the magnetization will probably have occurred after equilibration of the iron minerals (500–600°C, see above), and at a late stage of cooling from the temperature (say 500°C) required for amphibolite facies metamorphism.

A tentative scenario is outlined in Fig. 6. It is assumed that the country rocks were metamorphosed to amphibolite facies at about −1170 m.y. (the closest whole-rock Rb/Sr isochron) at which time, or somewhat later, the anorthosite was intruded and metamorphosed and the contact aureole formed. The U/Pb ages record the decay of metamorphism when the *WW* and *WZ* magnetizations were acquired. *WY* is acquired somewhat later, and is perhaps a little older than the whole-rock K/A and Rb/Sr mineral ages. *WX* may record even later cooling or perhaps a mild reheating event. If this scheme is approximately correct, then the h.c.f. 'hematite' magnetization was acquired between the youngest Rb/Sr whole-rock and the K/A ages, and the *WZ* magnetization corresponds to *about* the K/A ages. This is reasonable if one accepts Moorbath's (1967) figure of about 200°C as the temperature below which argon is retained. One simplifying feature of this model of the acquisition of magnetization in metamorphic rocks is that only physical processes are considered because the temperatures at which the magnetization was acquired are much lower than those of the chemical equilibration of the iron minerals (see above); the magnetizations are assumed to be t.r.m.'s acquired over long intervals and not c.r.m.'s. This is only a first attempt at a model for interpreting the ages of magnetization of metamorphic rocks and is obviously very very tentative.

The Grenville Polar Track

Poles from localities south of the Grenville Front are plotted in Fig. 7. Only poles obtained after demagnetization studies are given. The early results of Hood (1961) and DuBois (1962) are not shown for the reasons given by Irving *et al.* (1972). The poles in Figs. 7 and 11 are plotted in what is nowadays the North Atlantic because they are close to the observing area and both can be seen together. The more usual convention of plotting in the opposite hemisphere is used in the other figures.

The poles shaded in Fig. 7 apparently form a sequence which is referred to as the Grenville Track. The poles are as follows: Allard Lake (*A* Hargraves and Burt 1968) and Morin anorthosites h.c.f. magnetization (*M* Irving *et al*, 1974), Haliburton basic rocks (*H* Buchan and Dunlop 1973*a* and *b*), Ottawa diorites (*OT* Irving *et al.* 1972), Whitestone diorite (*WZ*), Wilberforce pyroxenite (*WP* Palmer and Carmichael 1973),

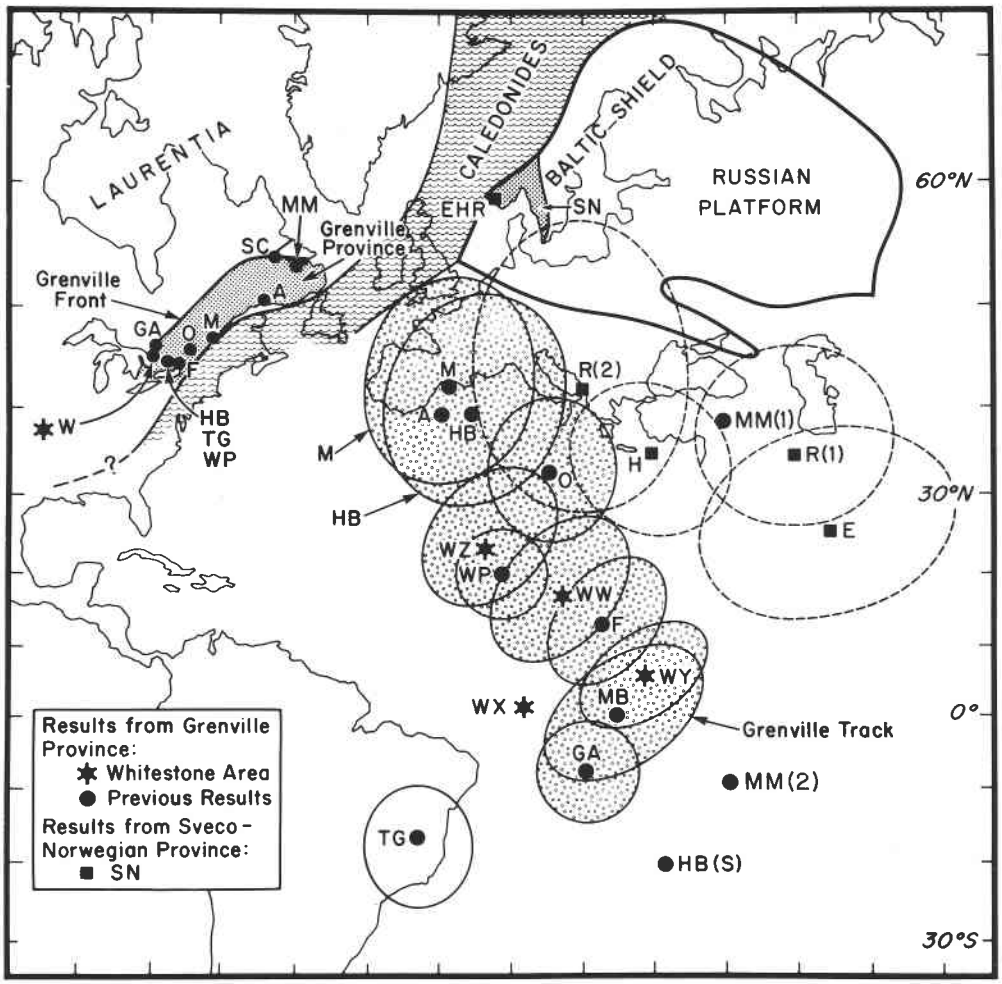


FIG. 7. The Grenville Polar Track and comparison with results from Norway. The error areas ($P = 0.05$) are marked. No errors are given for poles *WX* and *HB* (*S*) for which confidence limits cannot be reliably calculated. The error for *A* is large (20°) and has not been given to avoid confusion. The poles from the Grenville Track are shaded. The North Atlantic is closed and the Norwegian poles are rotated with the Baltic Shield using the rotation pole of Bullard *et al.* (1965). The Grenville Province and the Sveco-Norwegian province which are geologically similar (Wynne-Edwards and Hasan 1970) have similar ornament. The poles are itemized in the text.

Whitestone anorthosite with oxide segregations (*WW*), the post-tectonic unmetamorphosed Frontenac dikes (*F* Park and Irving 1972), Whitestone massive anorthosite (*WY*), the m.c.f. magnetization of the Morin anorthosite (*MB* Irving *et al.* 1974) and the anorthosite near the Grenville Front (*GA* Palmer and Carmichael 1973). The initial reason for defining the sense of this track was that the magnetization of the rock units involved had this age sequence as defined by cross-cutting and metamorphic relationships in the Grenville Province (Irving *et al.* 1972). In particular the Frontenac dikes (*F*) are unmetamorphosed and hence their magnetization is very probably younger than the highly metamorphic terrain into which they are intruded. Subsequent work on the Morin anorthosite has revealed two stable magnetizations, an h.c.f. magnetization of very high stability (*M*), and a m.c.f. magnetization (*MB*) with lower coercive force and blocking temperatures coexisting in the same specimens. Hence *M* very probably

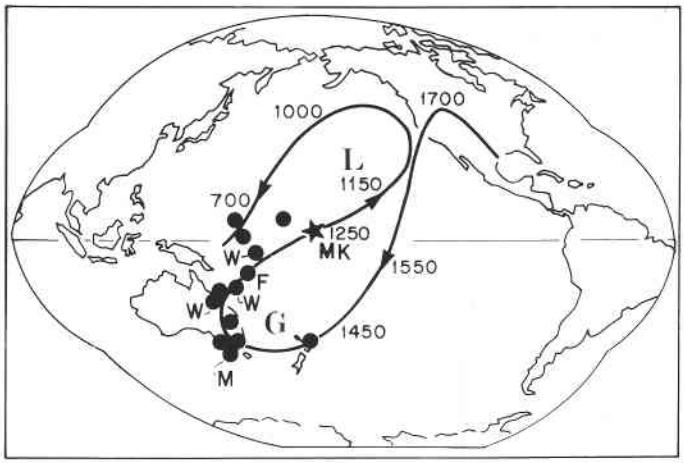


FIG. 8. Method (i) of reconstructing polar wander path for Laurentia by Fahrig *et al.* (1974). Poles from Grenville Province superimposed. The poles from the Whitestone area are marked *W*, *F* is from the Frontenac dikes and *M* is from the h.c.f. magnetization of the Morin anorthosite. *MK* is from rocks of the Mackenzie Igneous Episode. *G* are the Grenville and Logan Loops.

predates *MB* (Irving *et al.* 1974). Similarly in the present study the presence of a significant difference between the *WZ*, *WW* poles and the *WY* pole (Fig. 7) indicates that there is a time lapse between their magnetizations, and the higher blocking temperatures and higher coercivities of the former suggest that it is the older. This is a good argument because it is difficult to imagine a physical process by which a magnetization of lower blocking temperatures and coercivities survived a subsequent magnetization process that produced a magnetization of higher blocking temperature and higher coercivities. It is possible to invert the time order of these magnetizations by appealing to hypothetical chemical processes, but such arguments are difficult to sustain since the chemical equilibration of the iron minerals very probably occurred at temperatures much higher than those at which the magnetization was acquired (see above). With the exception of *F* all poles from the Grenville Track are from high-grade (high amphibolite or granulite) metamorphic terrain.

The age span of the Grenville Polar Track has been estimated as approximately -1120 (*M*) to -1000 (*GA*) m.y. (Irving *et al.* 1974). This is consistent with the age limits for the Whitestone poles given above. Apparently the Grenville Province moved from high latitudes to low latitudes during this interval (Fig. 7).

Certain poles from the Grenville Province do not fall on this Track. There are however good reasons why they should not. The Tudor Gabbro (pole *TG*, Fig. 7, Palmer and Carmichael 1973) is in the Hastings Basin which is an area of low-grade metamorphism (green schist facies). Hence the magnetization may be the primary t.r.m. acquired at the time of intrusion (estimated to be 1250 m.y.) and therefore older than the Grenville Track. The m.c.f. component (pole *HB*(*S*) Fig. 7) observed by Buchan and Dunlop (1973) in the Haliburton basic rocks falls beyond the younger end of the Grenville Track. Pole *HB* (*S*) is poorly defined and the errors ($\alpha_{95} = 16°$) may be underestimated because samples not sites have been given unit weight. Finally, recent work on the Mealy Mountain anorthosite from the eastern part of the Grenville Province gave two magnetizations *MM1* and *MM2* of Fig. 7 (Fahrig *et al.* 1974). *MM2* is probably not significantly different from poles *GA*, *HB* and *WY* of the Grenville Track, but *MM1* does appear to be to the east of the Track. The grade of metamorphism and the blocking temperatures of the magnetization are not known and hence the results are difficult to interpret. Fahrig *et al.* (1974) for example argue that *MM2* is the original t.r.m. acquired at the time of initial intrusion about -1500 m.y. ago so that this pole is in their view very much older than any from the Grenville Track.

The Grenville Polar Track is the first polar

path observed from metamorphic terrain and its geological significance is difficult to assess with confidence. The Track describes the motions of the southern part of the Grenville Province relative to the pole. The magnetic record has been imprinted during intervals of cooling in a manner which depends on the blocking temperatures of the magnetic minerals. Since the blocking temperatures can be measured, much information about the post-metamorphic thermal history can in principle be obtained. However, of more immediate interest is the kinematics of Precambrian continental motions, and in this context there are three pertinent questions. How is the Grenville Track connected to the polar path for the remainder of the Canadian Shield (referred to as Interior Laurentia)? What do these connections tell us about the motions of Grenville terrain relative to interior Laurentia? How does the Grenville Track relate to paleomagnetic results from the geologically comparable Sveco-Norwegian province of the Baltic Shield?

The Grenville Track and the Polar Path for Interior Laurentia

The ways in which the Grenville Track may be connected to the pole path of Interior Laurentia depend on the ages assigned to Grenville Poles. Three ways have been suggested. Other reconstructions are possible and we make no claim that our discussion is exhaustive.

(i) Grenville Track older than −1200 m.y.

Robertson and Fahrig (1971) suggested that in the polar path from Interior Laurentia there is a loop (Logan Loop *L* Fig. 8) extending in time from 1250 to 675 m.y., and the Grenville Track may be connected to the older end of this, so that the pole path forms a continuous letter *S* (Roy and Fahrig 1973; Fahrig *et al.* 1974). The new Whitestone poles and most of the other Grenville poles fall on the path. This reconstruction implies that the Whitestone (*W*) and Morin poles (*M*) are older than the Mackenzie intrusives, which are at least 1200 m.y. old and which may be substantially older (Gates and Hurley 1973). That is, their magnetization is older than the regional granulite and amphibolite metamorphism of Grenville terrain. This is improbable. Hence this method of linking Grenville poles to those from interior Laurentia is not considered likely.

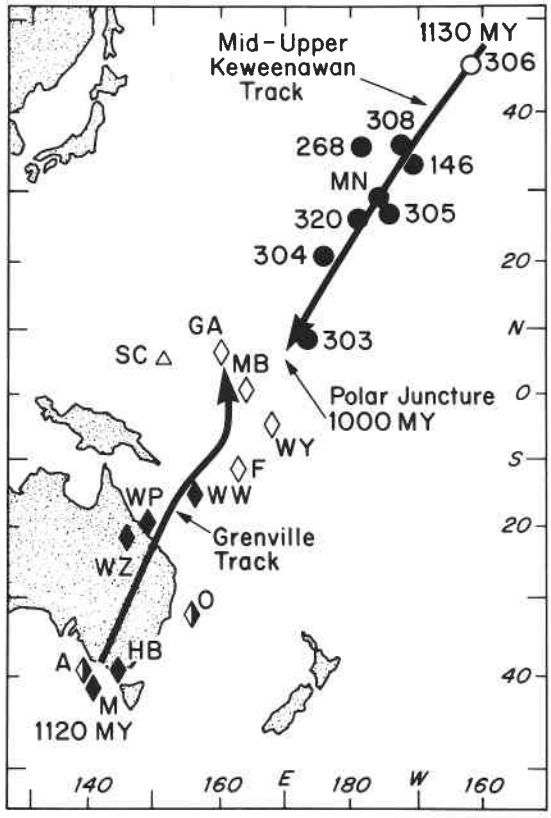


FIG. 9. Polar tracks from Grenville and Keweenawan rocks. The Grenville poles (diamonds) are identified in the text. The Keweenawan poles are as follows: *268* Robertson and Fahrig (1971), *303* (DuBois 1962; Vincenz and Yasukawa 1968), *304* (DuBois 1962; Vincenz and Yasukawa 1968), *305* (DuBois 1962; Palmer 1970), *146* (Palmer 1970), *306* (Books 1968; Palmer 1970), *308* (Beck 1970), *320* (Helsley and Spall 1972), *MN* (Robertson 1973), *SC* Seal-Croteau igneous rocks. The poles for the Tudor Gabbro and the less reliable poles from the Grenville Province are not plotted for the reasons given in the text.

(ii) Grenville Track −1125 to −1000 m.y.

Reconstructions of this type have been given by Buchan and Dunlop (1973) and Stewart and Irving (1974). The age assignment is based implicitly on cooling models of the type shown in Fig. 6. If this assignment is correct then the Grenville poles are the counterpart of poles from the contemporaneous Upper and Middle Keweenawan, and both are plotted in Fig. 9. The pole paths approach and appear to meet about −1000 m.y. This means that Grenvillia moved independently of Interior Laurentia and became sutured to it at that time, since polar junctures are

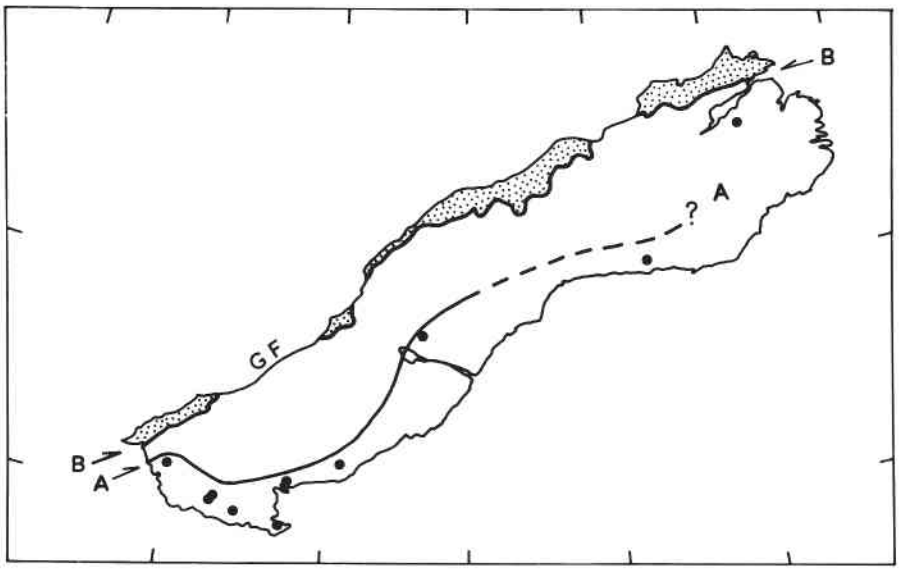


FIG. 10. Limits to the hypothetical Grenville suture. *BB* is the southern limit to which rocks from central Labrador and the Superior and Southern Provinces can be *firmly* traced into the Grenville Province. *AA* is the northern limit of localities from which poles (dots) of the Grenville Province have been obtained. The hypothetical suture lies between *AA* and *BB if* hypothesis (ii) is correct (Fig. 9).

the polar signature for continental collision and suturing. This occurred at the time of acquisition of the m.c.f. magnetization of the Morin anorthosite (*MB*), the massive Whitestone anorthosite (*WY*), and the Grenville front anorthosites (*GA*). It is of interest that the pole for the Seal-Croteau igneous rocks (Roy and Fahrig 1973) is also near the polar juncture and these rocks which are close to the Grenville Front (Fig. 7) may have therefore been magnetized at this time. From this standpoint it might be noted that the prominent metamorphic episode in the Grenville Province at −1100 m.y. and the maximum development of Keweenawan volcanism (Duluth Gabbro *308*, North Shore Volcanics *306*, Mamainse Lavas *MN* etc., Fig. 9) was not contemporaneous with the suturing, but occurred 100 m.y. previously at a time when Grenvillia was situated 3000 km to the SE of interior Laurentia. Suturing was essentially contemporaneous with the K/A ages, development of m.c.f. magnetizations and the intrusion of the Frontenac dikes. A further consequence of this method of connecting poles from the Grenville Province to the polar track for Interior Laurentia is that the suture must lie to the north of those localities from which poles of the Grenville Track have been obtained (*AA* of Fig. 10). The northern limit to the suture is the southern limit

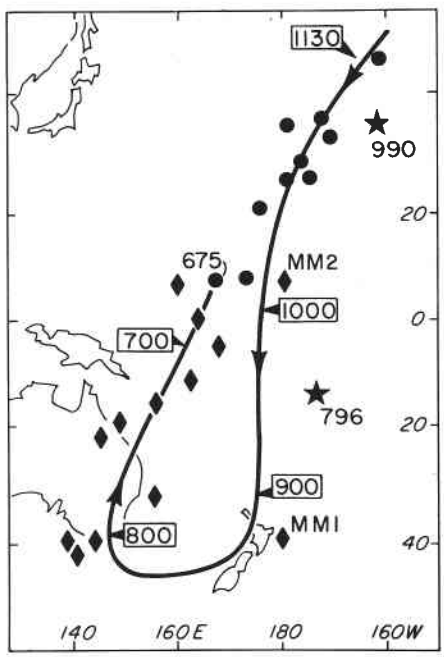


FIG. 11. Method (iii) of linking Grenville poles to those from Interior Laurentia. The poles *990* (Stoer Group, Stewart and Irving 1974) and *796* (Torridon Group, Irving and Runcorn 1957) are from western Scotland and have been corrected for closure of the Atlantic. Their numbers are their ages in m.y. obtained from Rb/Sr whole-rock isochrons.

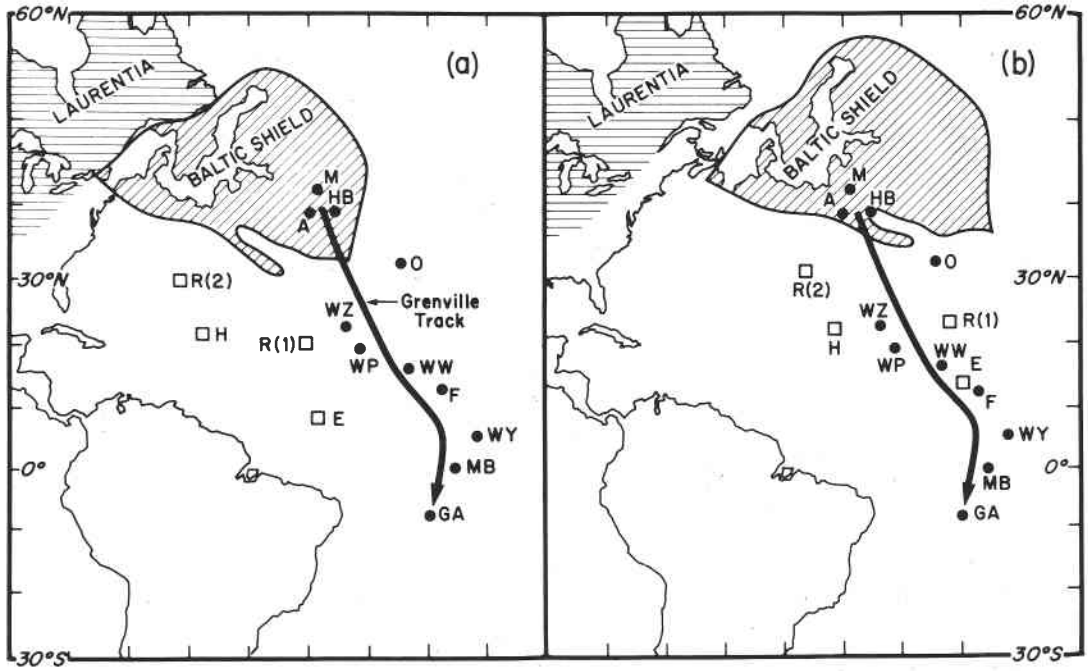


FIG. 12. Possible relative positions of Laurentia and the Baltic Shield between about −1000 and −600 m.y. For explanation see text.

to which rocks of central Labrador and the Southern and Superior Provinces can be traced with certainty into the Grenville Province (*BB* of Fig. 10). There is no geological evidence for a suture between *AA* and *BB* but the geological signatures of sutures at such deep crustal levels (10 to 20 km) may be quite obscure.

(iii) Grenville Track younger than −1000 m.y.

The Keweenawan polar track may loop into the southern hemisphere, pass through the Grenville poles, and return to the Franklin pole (675 m.y.) as shown in Fig. 11. Since the path must first move far to the south in order to preserve the apparent time order of Grenville poles, the age of the latter must be substantially less than −1000 m.y., probably less than −900 m.y., and the youngest must be as young as −700 m.y. Reconstructions of this type have been made by Stewart and Irving (1974). They imply that the h.c.f. magnetization is younger than the K/A ages of Grenville rocks by 100 m.y. or more. It has been argued above that the magnetization was acquired in the temperature range 200–300°C. Thus the reconstruction of Fig. 11 implies that the K/A ages in fact reflect much higher temperatures than the 200°C normally accepted. It is also noteworthy that other poles in this time range from Laurentia (*796* and *909* of Fig. 11) are far from this curve; the *990* pole is 50° or more away from its expected position.

In summary therefore it may be said that (i) is unlikely, and (iii) is possible. Option (ii) is also possible, but it implies very large motions of Grenville relative to Shield, and a suture for which there appears at present to be no geological evidence.

Sveco-Norwegian Poles Compared

In Fig. 7 the poles from Sveco-Norwegian Province of the Baltic Shield are plotted after correction for closure of the North Atlantic. Four poles are available: *E* Egersund dolerites (Storetvedt and Gidskehaug 1968; Poorter 1972), *H* Hallenden dikes (Poorter 1972) and two results from the Rogaland metamorphic basement *R1* (Poorter 1972) and *R2* (Hargraves and Fish 1972). *R1* is a result from an extensive sampling of many rock types including the Egersund anorthosite (19 sites). *R2* is from the

Egersund anorthosite (5 sites). Hargraves and Fish (1972) show that pole *R2* is not significantly different from the pole for the Allard Lake anorthosite (*A*) of Hargraves and Burt (1968), but both error circles are very large and the comparison cannot reveal differences of less than 20° or so. The other Sveco-Norwegian poles are however systematically displaced to the east, for poles *R1* and *E* the displacement is significant despite the large error circles. We attribute this discrepancy to the Caledonian orogenic cycle which occurred subsequently to the Grenville orogeny.

A feature of the Caledonide fold belt is the similar shape of the tectonic fronts in SE Canada and Norway. In Fig. 12(a) the fronts are fitted together, the Baltic Shield being placed immediately SE of Laurentia. This brings the Sveco-Norwegian poles through and beyond the Grenville Track. A better fit of poles is achieved by placing the Baltic Shield somewhat further to the east (Fig. 12(b)). This result is consistent with either of two hypotheses. Firstly, the Caledonian orogen was always a comparatively narrow trough and the margins moved by transcurrent motion from the position of Fig. 12(b) to that of Fig. 7. Secondly, the Caledonian orogen was the site of a wide plate-style ocean but that during the compressional phases the margins returned to approximately the same relative positions. Clearly many more observations are needed before a proper assessment can be made but it is clear that *broad* agreement between results from the Grenvillian and Sveco-Norwegian terrain implies that prior to the Caledonian orogenic cycle these two regions were near together.

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